

Galaxy Tools 2.0

Bring your own!

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Why can't users install tools ?

- We don't want to limit what you can do!
 - I need “xyz” and it's not on the server I use, help please?
- You can already run code with RStudio and Jupyter*
- We can isolate storage and compute

Why can't users install tools ?

Templating with Cheetah

```
<tool id="cat" version="0.1">
  <description>tail-to-head</description>
  <requirements>
    <requirement type="container">busybox</requirement>
  </requirements>
  <command><![CDATA[
cat
#for dataset in datasets:
    '$dataset'
#end for
> '$output1'
  ]]></command>
  <inputs>
    <input name="datasets" format="data" type="data" multiple="true"/>
  </inputs>
  <outputs>
    <output name="output1" format_source="datasets" />
  </outputs>
</tool>
```

Templating with Cheetah

```
<command><![CDATA[  
    #from pathlib import Path  
    #user_id = $__app__.model.session().query($__app__.model.User.id).one()  
    #open(f"{Path.home()}/a_file", "w").write("Hello!")  
]]></command>
```

How do we manage 1000s of tools ?

The screenshot shows the GitHub interface for the `galaxyproject / tools-iuc` repository. The top navigation bar includes the repository name, a search bar, and icons for repository management. Below this, a secondary navigation bar shows tabs for `Code`, `Issues` (436), `Pull requests` (337), `Discussions`, `Actions`, `Projects`, `Security`, and `Insights`. The `Pull requests` tab is selected and underlined.

Below the navigation bar, there is a section for filters and a search bar. The search bar contains the text `is:pr is:open`. To the right of the search bar are buttons for `Labels` (24) and `Milestones` (1), and a green button labeled `New pull request`.

The main content area displays a list of pull requests. Each entry includes a checkbox, a pull request icon, a title, a status indicator (checkmark or cross), and additional information such as the number of tasks, the reviewer, and the time since it was opened.

☐		337 Open	✓	5,450 Closed	Author	Label	Projects	Milestones	Reviews	Assignee	Sort
☐		updates to samtools 1.22, adds report feature for ampliconclip	•		#7075 opened 20 hours ago by bwlang	Review required	2 of 5 tasks				
☐		Updating tools/ucsc_tools/fatovcf from version 473 to 482	✗		#7073 opened yesterday by gxydevbot	Review required					
☐		Updating tools/tasmanian_mismatch from version 1.0.7 to 1.0.9	✓		#7071 opened yesterday by gxydevbot	Review required					
☐		Upgrade samtools dependencies	✗		#7068 opened 4 days ago by bwlang	Review required	3 of 5 tasks				2
☐		New tool: PhyloPhlAn an integrated pipeline for phylogenetic profiling of genomes and metagenomes	✗		#7067 opened 5 days ago by neo417	Draft	3 of 5 tasks				

There has to be a better way!

Javascript expressions & JSON inputs

```
class: GalaxyUserTool
id: cat_user_defined
version: "0.1"
name: Concatenate Files
description: tail-to-head
container: busybox
shell_command: |
  cat $(inputs.datasets.map((input) => input.path).join(' ')) > output.txt
inputs:
  - name: datasets
    multiple: true
    type: data
outputs:
  - name: output1
    type: data
    format_source: datasets
    from_work_dir: output.txt
```


Javascript expressions & JSON inputs

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    multiple: true
    type: data
outputs:
  - name: output1
    type: data
    format_source: datasets
    from_work_dir: output.txt
```

```
var inputs = {
  "datasets": [
    {
      "class": "File",
      "location": "step_input://1",
      "format": "csv",
      "path": "/Users/mvandenb/src/galaxy/databa
      "basename": "markers.csv",
      "nameroot": "markers",
      "nameext": ".csv"
    }
  ],
  "chromInfo": "/tmp/shared/ucsc/chrom/?..len",
  "dbkey": "?",
  "__input_ext": "input"
};
```

Javascript expressions & JSON inputs

Tool Editor

```
1  class: GalaxyUserTool
2  id: cat_user_defined
3  version: "0.1"
4  name: Concatenate Files
5  description: tail-to-head
6  container: busybox
7  shell_command: |
8  | cat $(inputs.datasets.map((input) => input.path).join(' ')) > output.txt
9  inputs:
10  | - name: datasets
11  |   multiple: true
12  |   type: data
13  outputs:
14  | - name: output1
15  |   type: data
16  |   format_source: datasets
17  |   from_work_dir: output.txt|
```

```
(parameter) input: {
  readonly class: "File";
  readonly basename: string;
  readonly location: string;
  readonly path: string;
  readonly listing: readonly string[] | null;
  readonly nameroot: string | null;
  readonly nameext: string | null;
  readonly checksum: string | null;
  readonly size: number;
}
```

Javascript expressions & JSON inputs

Tool Editor

```
1  class: GalaxyUserTool
2  id: cat_user_defined
3  version: "0.1"
4  name: Concatenate Files
5  description: tail-to-head
6  container: busybox
7  shell_command: |
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  readonly nameext: string | null;
  readonly checksum: string | null;
  readonly size: number;
}
```

Monaco + YAML schemas

Tool Editor

Save

```
1 class: GalaxyUserTool
2 id: cat_user_defined
3 version: "0.1"
4 name: Concatenate Files
5 description: tail-to-head
6 container: busybox
7 shell_command: |
8   cat $(inputs.datasets.map((input) => input.path).join(' ')) > output.txt
9 inputs:
10   - name: datasets
11     multiple: true
12     type: data
13 outputs:
14   - name: output1
15     type: data
16     format_source: datasets
17     from_work_dir: output.txt
```

(property) path: `string`

Path

@description — The absolute path to the file on disk.

Monaco + Typescript interfaces

Tool Editor

```
1 class: GalaxyUs
2 id: cat_user_de
3 version: "0.1"
4 name: Concatena
5 description: ta
6 container: busy
7 shell_command:
8   cat $(inputs.datasets.map((input) => input.path).join(' ')) > output.txt
9 inputs:
10   - name: datasets
11     multiple: true
12     type: data
13 outputs:
14   - name: output1
15     type: data
16     format_source: datasets
17     from_work_dir: output.txt
```

Property 'datasets' does not exist on type '{ readonly datasets: readonly { readonly class: "File"; readonly basename: string; readonly location: string; readonly path: string; readonly listing: readonly string[] | null; readonly nameroot: string | null; readonly nameext: string | null; readonly checksum: string | null; readonly size: number; }[]; }'. Did you mean 'datsets'?

any

[View Problem \(\F8\)](#) No quick fixes available

Save

Behind the scenes

- Tool parameters are modeled as Pydantic models
 - As a general schema that describes “A Galaxy Tool”
 - As a specialized schema for “This Galaxy Tools’ Potential Input Object”
- Pydantic models are transform to JSON schema
- JSON schema is transformed to typescript interface for runtime inputs
- Monaco Editor renders tool source
 - Knows what to do with yaml schema for tool
 - Understands typescript interface and compares to code in javascript fragments

Behind the scenes

```
"inputs": {  
  "additionalProperties": false,  
  "properties": {  
    "datasets": {  
      "items": {  
        "$ref": "#/components/schemas/DataInternalJson"  
      },  
      "title": "Datasets",  
      "type": "array"  
    }  
  },  
  "required": [  
    "datasets"  
  ],  
}
```



```
"inputs": {
  "additionalProperties": false,
  "properties": {
    "datasets": {
      "items": {
        "$ref": "#/components/schemas/DataInternalJson"
      },
      "title": "Datasets",
      "type": "array"
    }
  },
  "required": [
    "datasets"
  ],
```

```
"DataInternalJson": {
  "additionalProperties": false,
  "properties": {
    "class": {
      "const": "File",
      "title": "Class",
      "type": "string"
    },
    "basename": {
      "description": "The base name of the file, that is, the name",
      "title": "Basename",
      "type": "string"
    },
    "location": {
      "title": "Location",
      "type": "string"
    },
    "path": {
      "description": "The absolute path to the file on disk.",
      "title": "Path",
      "type": "string"
    }
  }
}
```

Enabling User-Defined Tools

To enable this feature:

1. Set `enable_beta_tool_formats: true` in your Galaxy configuration.
2. Create a role of type `Custom Tool Execution` in the admin user interface.
3. Assign users or groups to this role.

Sharing User-Defined Tools

User-defined tools are private to their creators. However, if a tool is embedded in a workflow, any user who imports that workflow will automatically have the tool created in their account.

These tools can also be exported to disk and loaded like regular tools, enabling instance-wide availability if needed.

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Limitations

The user-defined tool language is still evolving, and additional safety audits are ongoing.

Current limitations include:

- [configfiles](#) are not supported
- Access to reference data is not supported
- Access to metadata and metadata files (such as BAM indexes) is not supported
- Access to the `extra_files` directory is not supported

Coming to a server near you soon!

The screenshot displays the Galaxy web interface. On the left is a sidebar with navigation links: Home, Tools, Workflows, Workflow Invocations, Custom Tools (highlighted with a wrench icon), Visualization, and Histories. The main content area is split into two panels. The left panel, titled 'Custom Tools', shows a tool card for 'Convert h5ad to CSV table' with a sub-description 'Generates a table', a timestamp 'about 12 hours ago', a version '0.6', and an 'Edit' button. Below the card, it says 'All items loaded'. The right panel, titled 'Tool Editor', shows a code editor with the following YAML configuration:

```
1 class: GalaxyUserTool
2 id: h5ad-to-csv
3 version: "0.6"
4 name: Convert h5ad to CSV table
5 description: Generates a table
6 container: afgane/bioc-galaxy:3.21.live.2
7 shell_command: ln -s $(inputs.input1.path) input
8 inputs:
9   - name: script
10     type: data
11   - name: input1
12     type: data
13     extensions:
14       - h5ad
15 outputs:
16   - name: output1
17     type: data
18     format: csv
19     from_work_dir: spatial_expression_data.csv
20
```

A 'Save' button is visible in the top right corner of the Tool Editor panel.

Special Thanks to

John Chilton
Michael Crusoe
Nicola Soranzo